



Network Reference Model



Foreword

- In the digital era, various information is presented as data in our life. What is data? How is data transmitted?
- In this course, we will use the network reference model to understand the "life" of data.



Objectives

- On completion of this course, you will be able to:
 - Understand the data definition and transmission process.
 - Understand the concepts and advantages of the network reference model.
 - Understand common standard protocols.
 - Understand the data encapsulation and decapsulation processes.



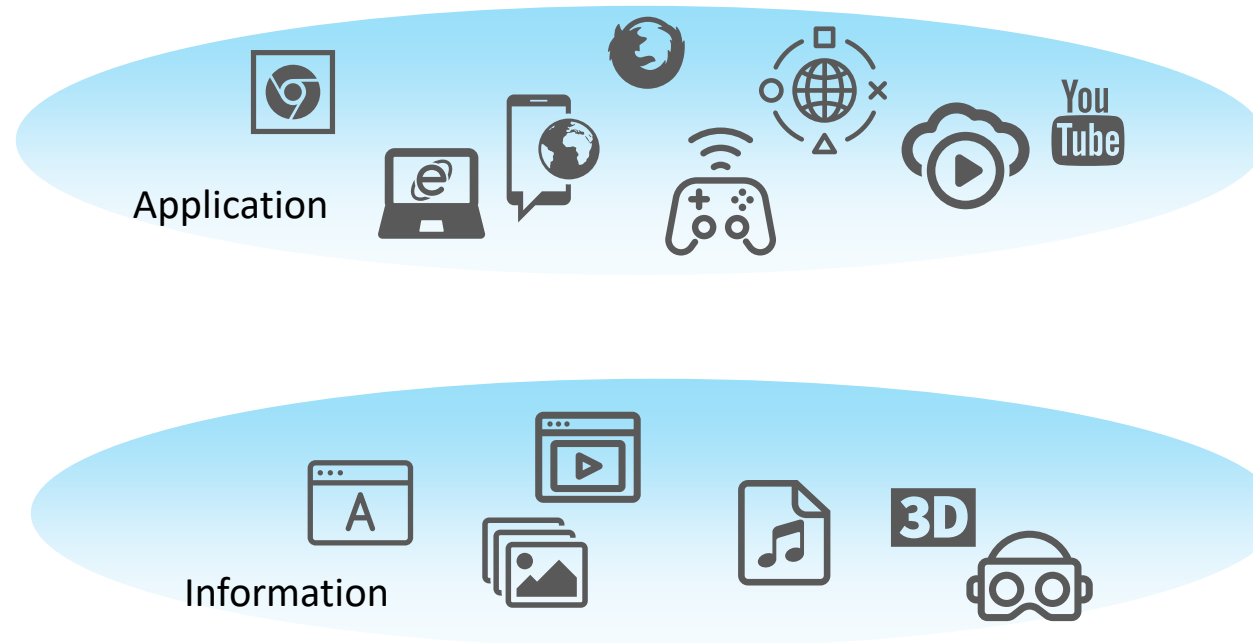
Contents

- 1. Applications and Data**
2. Network Reference Model and Standard Protocols
3. Data Communication Process



Origin of the Story - Applications

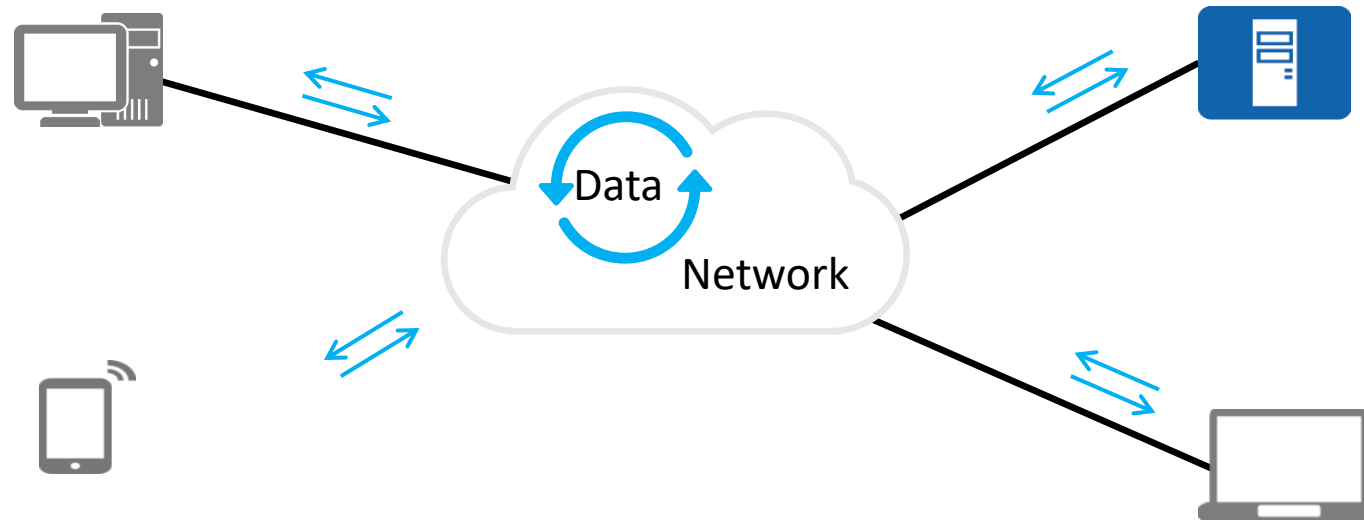
- Applications are used to meet various requirements of people, such as web page access, online gaming, and online video playback.
- Information is generated along with applications. Texts, pictures, and videos are all information presentation modes.





Application Implementation - Data

- Data generation
 - In the computer field, data is the carrier of all kinds of information.
- Data transmission
 - Data generated by most applications needs to be transmitted between devices.



Does an application need to complete the entire process from data generation to data transmission?



Contents

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- 2. Network Reference Model and Standard Protocols**
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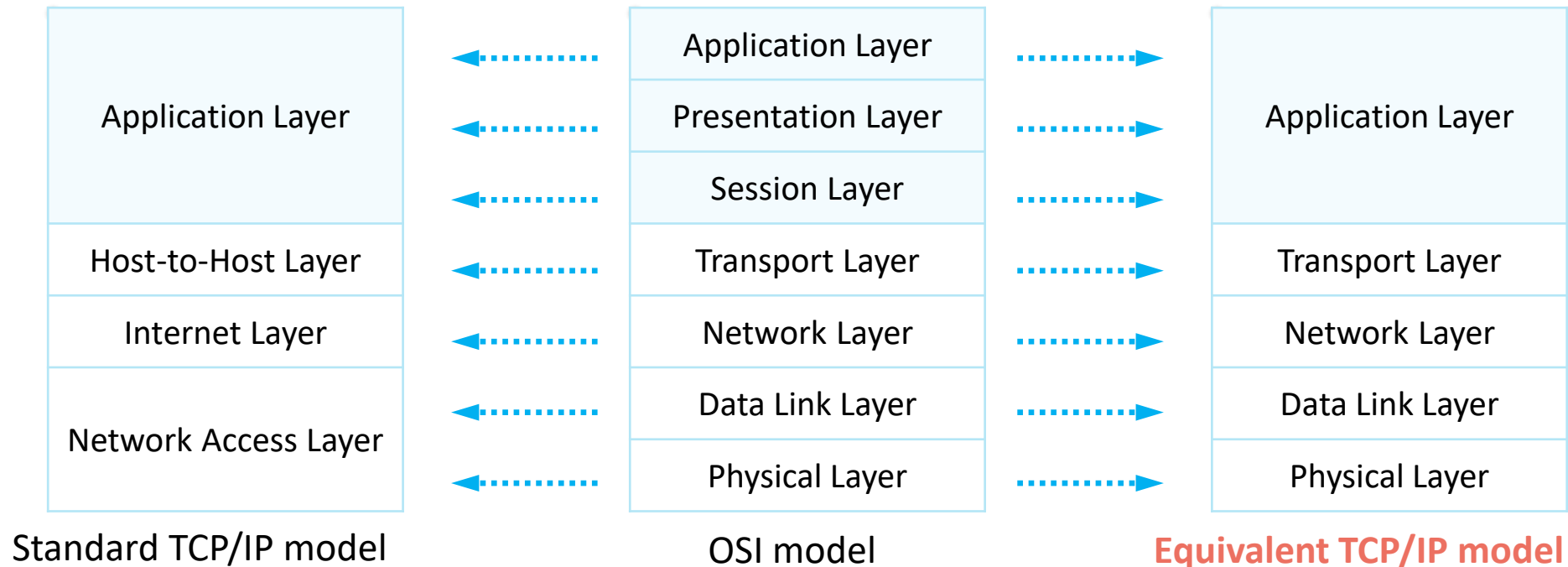
OSI Reference Model

7. Application Layer	Provides interfaces for applications.
6. Presentation Layer	Translates data formats to ensure that the application-layer data of one system can be identified by the application layer of another system.
5. Session Layer	Establishes, manages, and terminates sessions between communicating parties.
4. Transport Layer	Establishes, maintains, and cancels an end-to-end data transmission process; controls transmission speeds and adjusts data sequences.
3. Network Layer	Defines logical addresses and transfers data from sources to destinations.
2. Data Link Layer	Encapsulates packets into frames, transmits frames in P2P or P2MP mode, and implements error checking.
1. Physical Layer	Transmits bitstreams over transmission media and defines electrical and physical specifications.



TCP/IP Reference Model

- The OSI protocol stack is complex, and the TCP and IP protocols are widely used in the industry. Therefore, the TCP/IP reference model becomes the mainstream reference model of the Internet.





Common TCP/IP Protocols

- The TCP/IP protocol stack defines a series of standard protocols.

Application Layer	Telnet	FTP	TFTP	SNMP
	HTTP	SMTP	DNS	DHCP
Transport Layer	TCP		UDP	
Network Layer	ICMP		IGMP	
	IP			
Data Link Layer	PPPoE			
	Ethernet		PPP	
Physical Layer	...			



Common Protocol Standardization Organizations

- Internet Engineering Task Force (IETF)
 - IETF is a voluntary organization responsible for developing and promoting Internet protocols (especially protocols that constitute the TCP/IP protocol suite), and releasing new or replacing old protocol standards through RFCs.
- Institute of Electrical and Electronics Engineers (IEEE)
 - IEEE has formulated about 30% of standards in the electronics, electrical, and computer science fields worldwide. Those standards include well-known IEEE802.3 (Ethernet) and IEEE802.11 (Wi-Fi).
- International Organization for Standardization (ISO)
 - ISO is an international organization that plays an important role in the formulation of computer network standards, such as the OSI model defined in ISO/IEC 7498-1.



Application Layer

Application
Layer

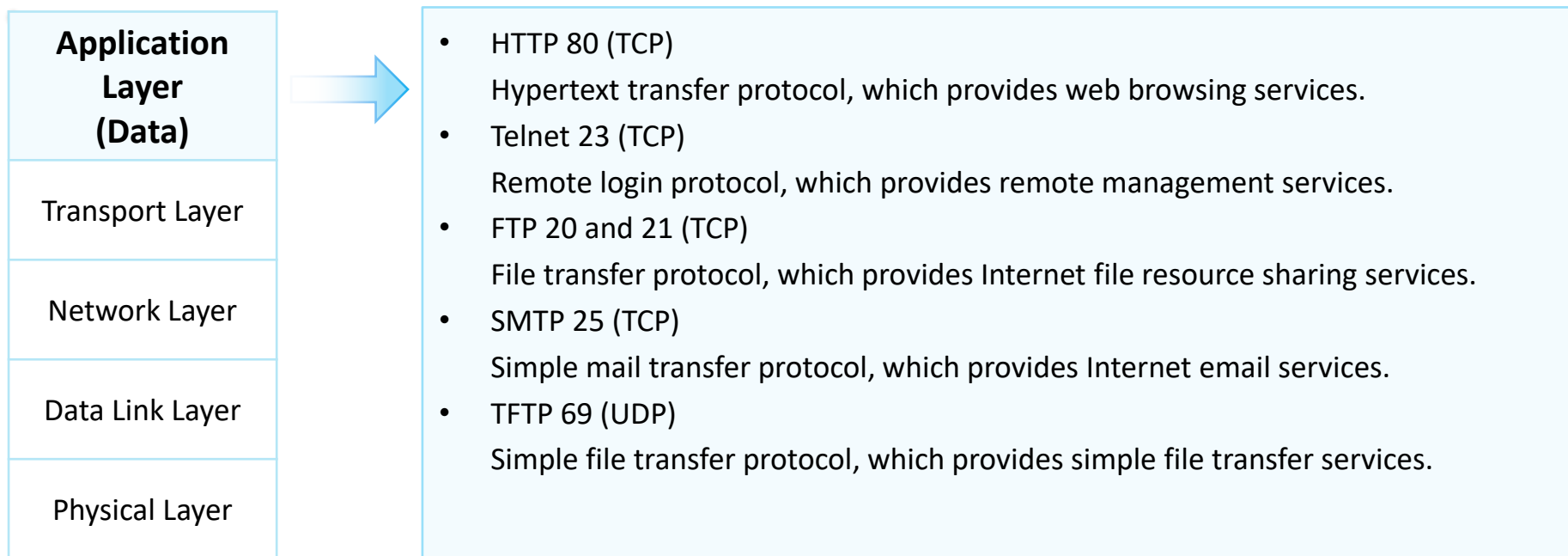
Transport
Layer

Network
Layer

Data Link
Layer

Physical
Layer

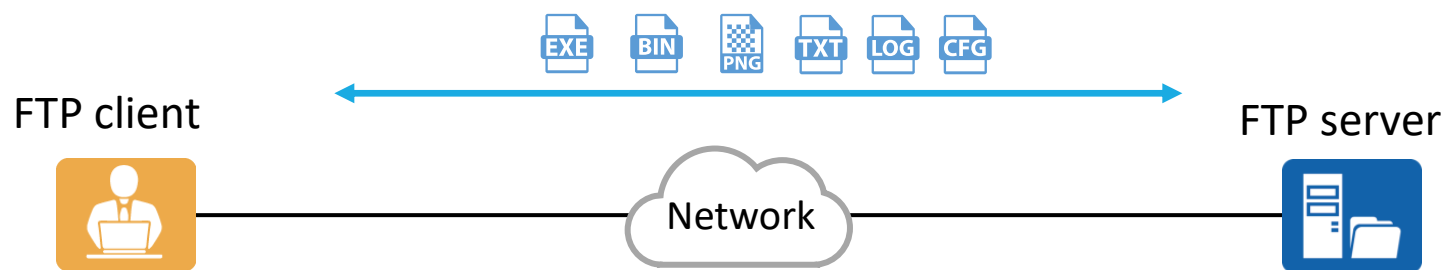
- The application layer provides interfaces for application software so that applications can use network services. The application layer protocol designates transport layer protocols and ports.
- PDUs transmitted at the network layer are called data.





Common Application Layer Protocols - FTP

- The File Transfer Protocol (FTP) transfers files from one host to another to implement file download and upload. This protocol adopts the client/server (C/S) structure.

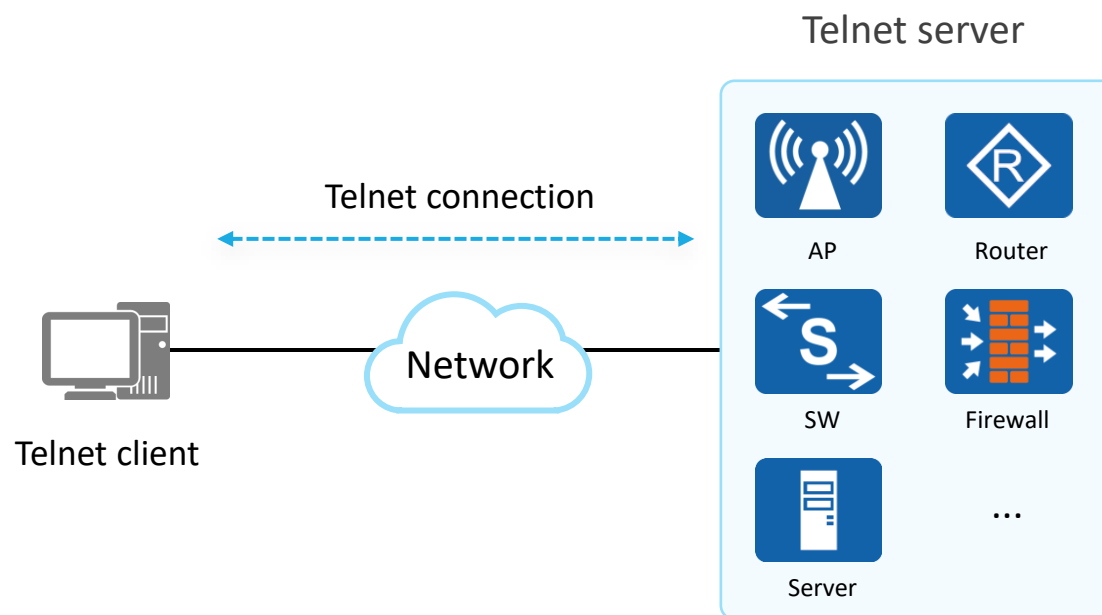


FTP client: provides commands for local users to operate files on a remote server. A user can install an FTP client program on a PC and set up a connection with an FTP server to operate files on the server.

FTP server: a device that runs the FTP service. It provides the access and operation functions for remote clients, allowing users to access the FTP server through the FTP client program and access files on the server.

Common Application Layer Protocols - Telnet

- Telnet is a standard protocol that provides remote login services on a network. It provides users with the ability to operate remote devices through local PCs.

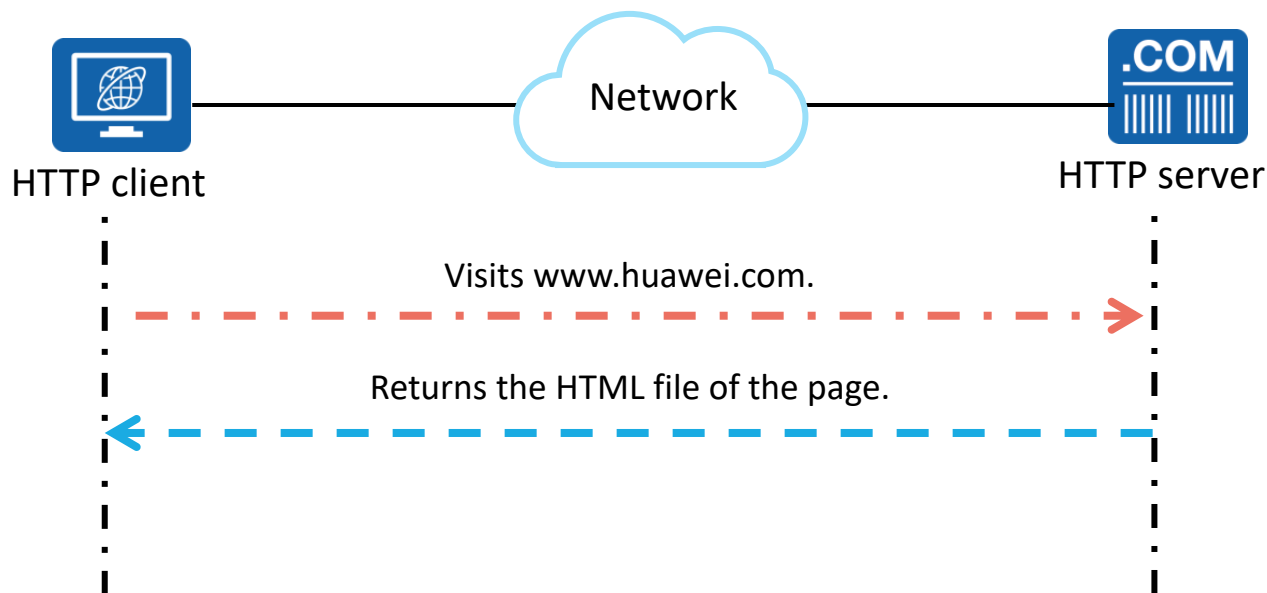


A user connects to a Telnet server through the Telnet client program. The commands entered on the Telnet client are executed on the server, as if the commands were entered on the console of the server.



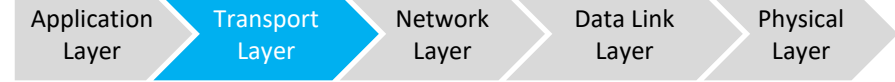
Common Application Layer Protocols - HTTP

- Hypertext Transfer Protocol (HTTP): is one of the most widely used network protocols on the Internet. HTTP was originally designed to provide a method for publishing and receiving HTML pages.

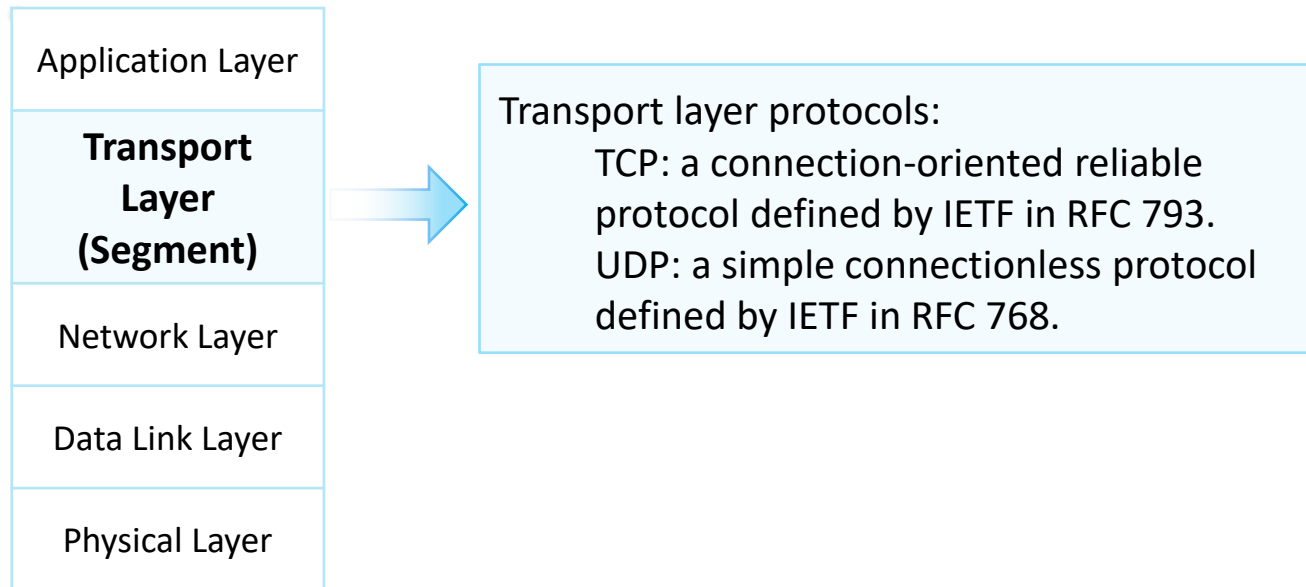




Transport Layer



- A transport layer protocol receives data from an application layer protocol, encapsulates the data with the corresponding transport layer protocol header, and helps establish an end-to-end (port-to-port) connection.
- PDUs transmitted at the transport layer are called segments.





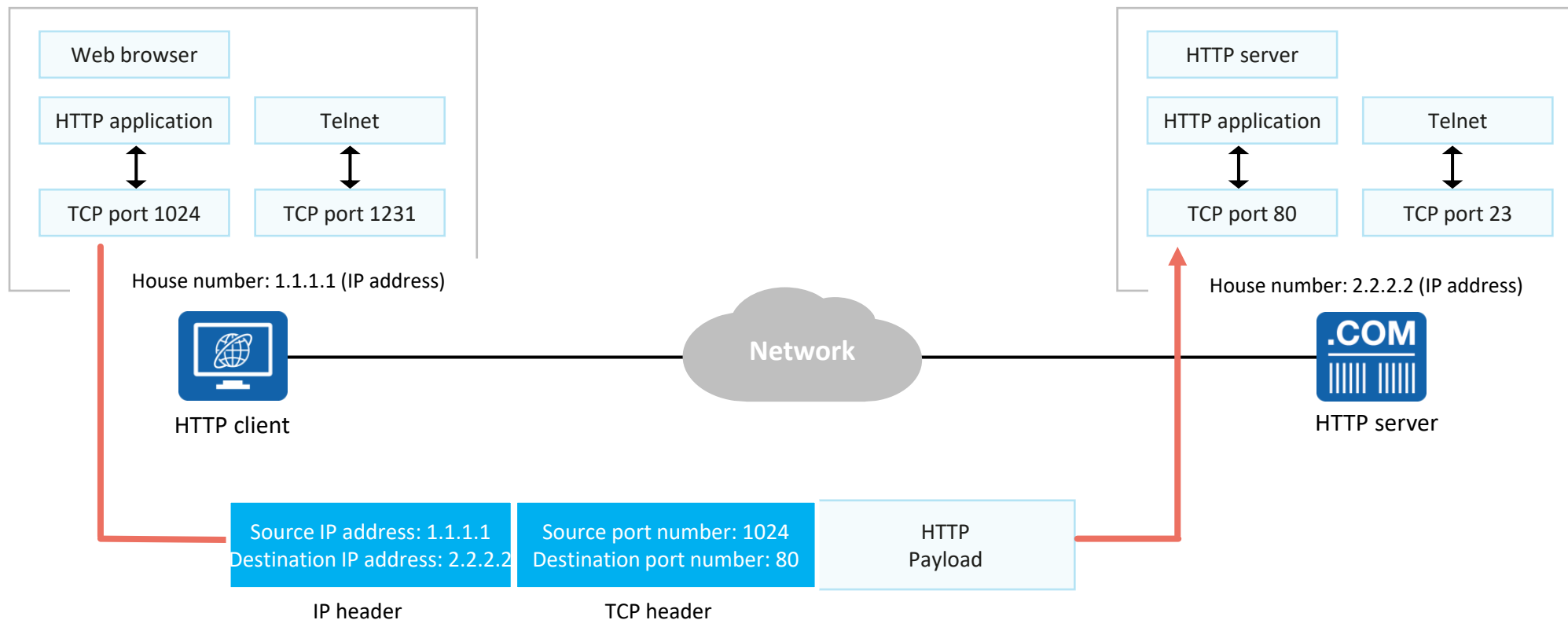
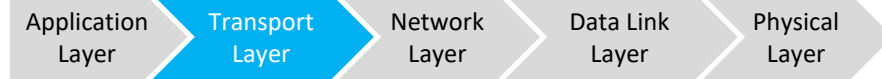
TCP and UDP - Header Formats

Source Port (16)			Destination Port (16)		TCP header 20 bytes
Sequence Number (32)					
Acknowledgement Number (32)					
Header Length (4)	Reserved (6)	Control Bits (6)	Window (16)		
Checksum (16)			Urgent (16)		
Options					
Data (varies)					

Source Port (16)	Destination Port (16)	} UDP header 8 bytes
Length (16)	Checksum (16)	
Data (if any)		



TCP and UDP - Port Numbers

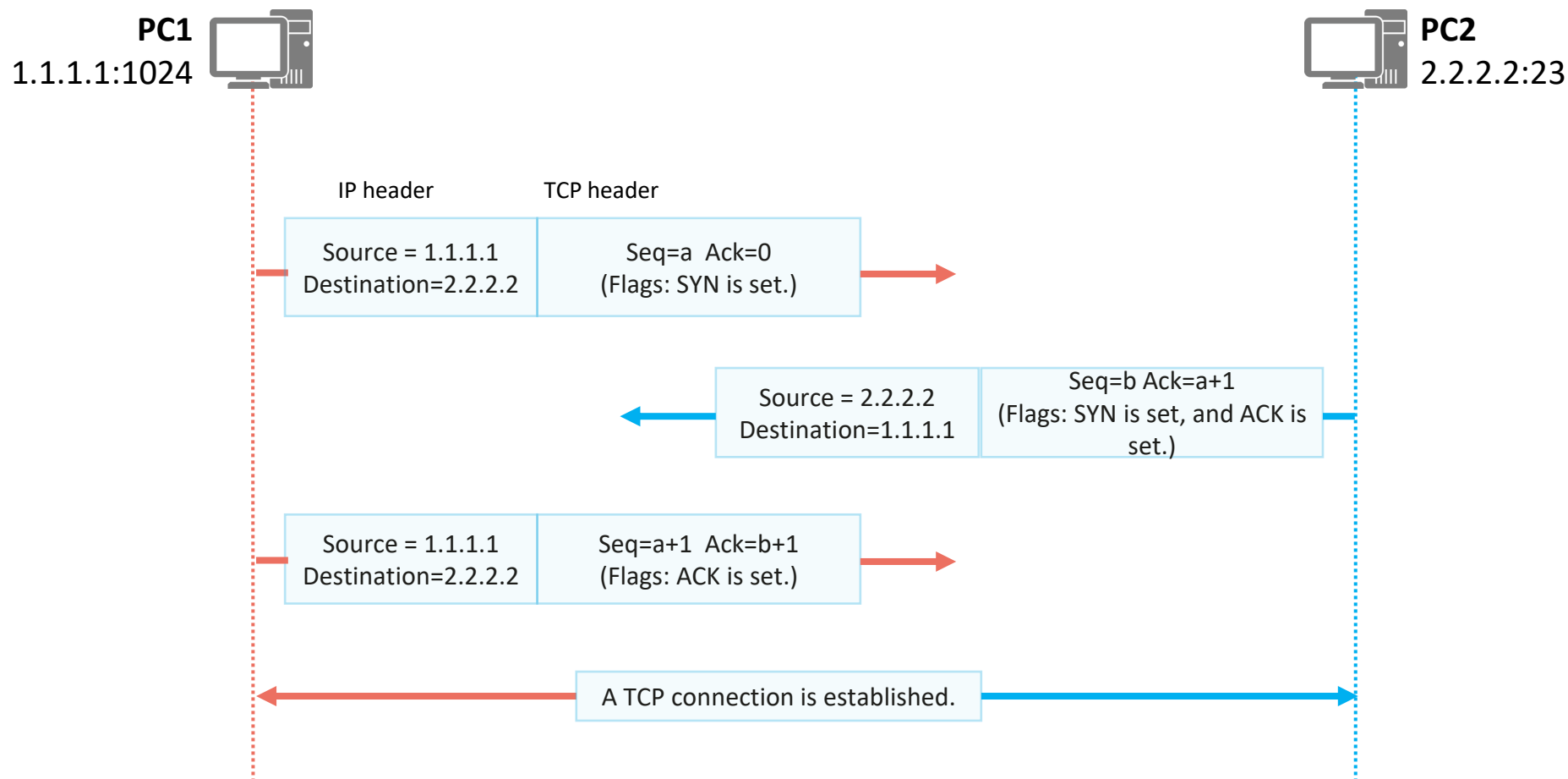


- Generally, the source port used by a client is randomly allocated, and the destination port is specified by the application of a server.
- The system generally selects a source port number that is greater than 1023 and is not being used.
- The destination port number is the listening port of the application (service) enabled on the server. For example, the default port number for HTTP is 80.



TCP Connection Setup - Three-Way Handshake

- Before sending data, a TCP-based application needs to establish a connection through three-way handshake.





TCP Sequence Number and Acknowledgment Number

Application Layer

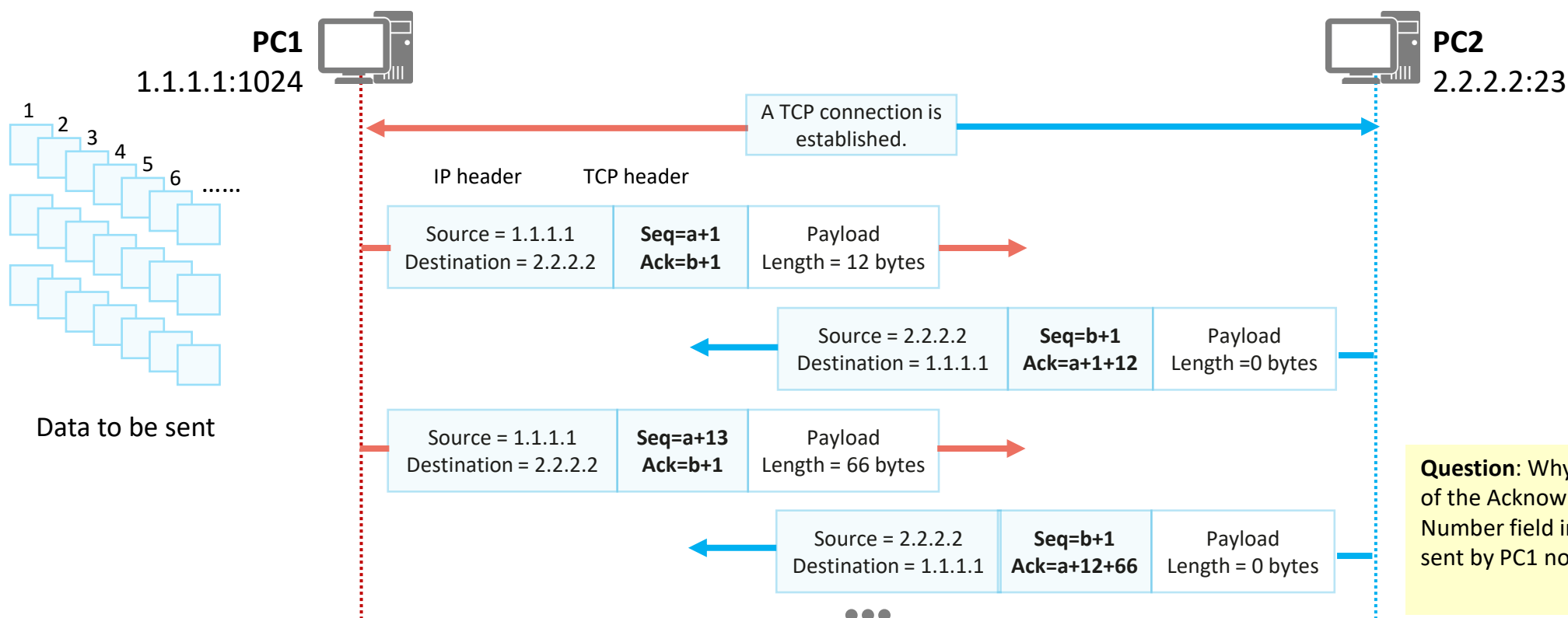
Transport Layer

Network Layer

Data Link Layer

Physical Layer

- TCP uses the Sequence Number and Acknowledgment Number fields to implement reliable and ordered data transmission.





TCP Window Sliding Mechanism

Application Layer

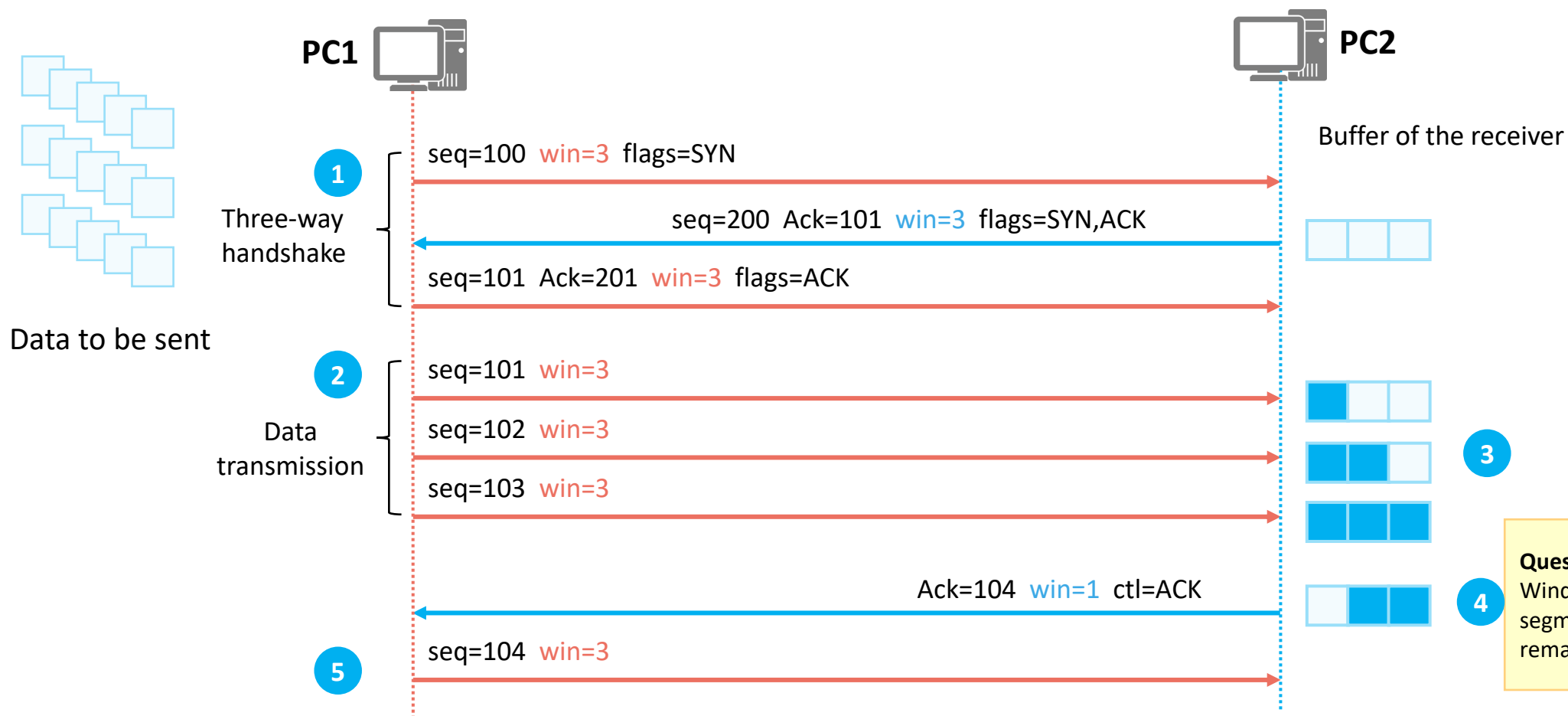
Transport Layer

Network Layer

Data Link Layer

Physical Layer

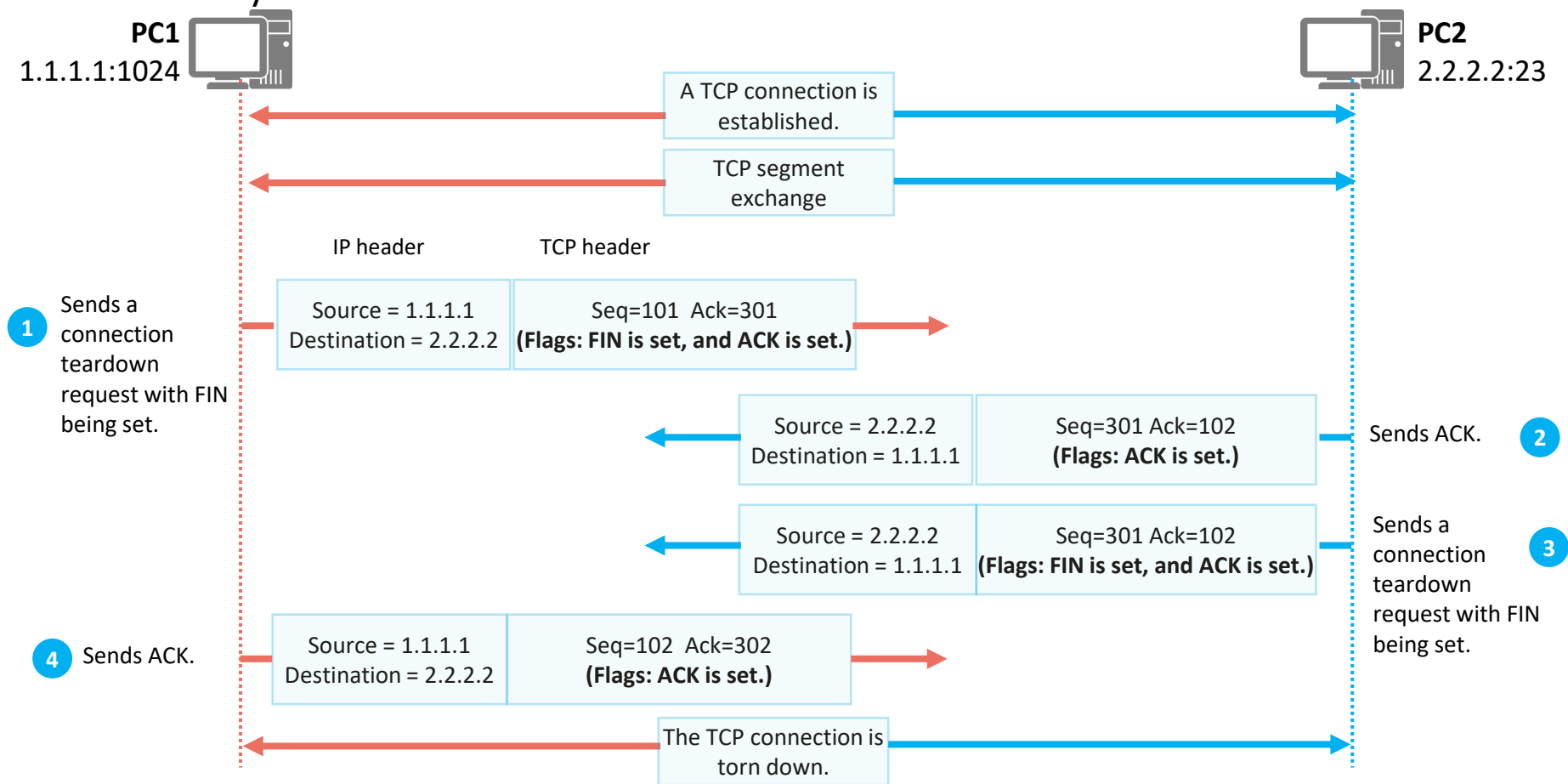
- TCP uses the sliding window mechanism to control the data transmission rate.





TCP Shutdown - Four-Way Handshake

- After data transmission is complete, TCP needs to use the four-way handshake mechanism to disconnect the TCP connection and release system resources.





Network Layer

Application
Layer

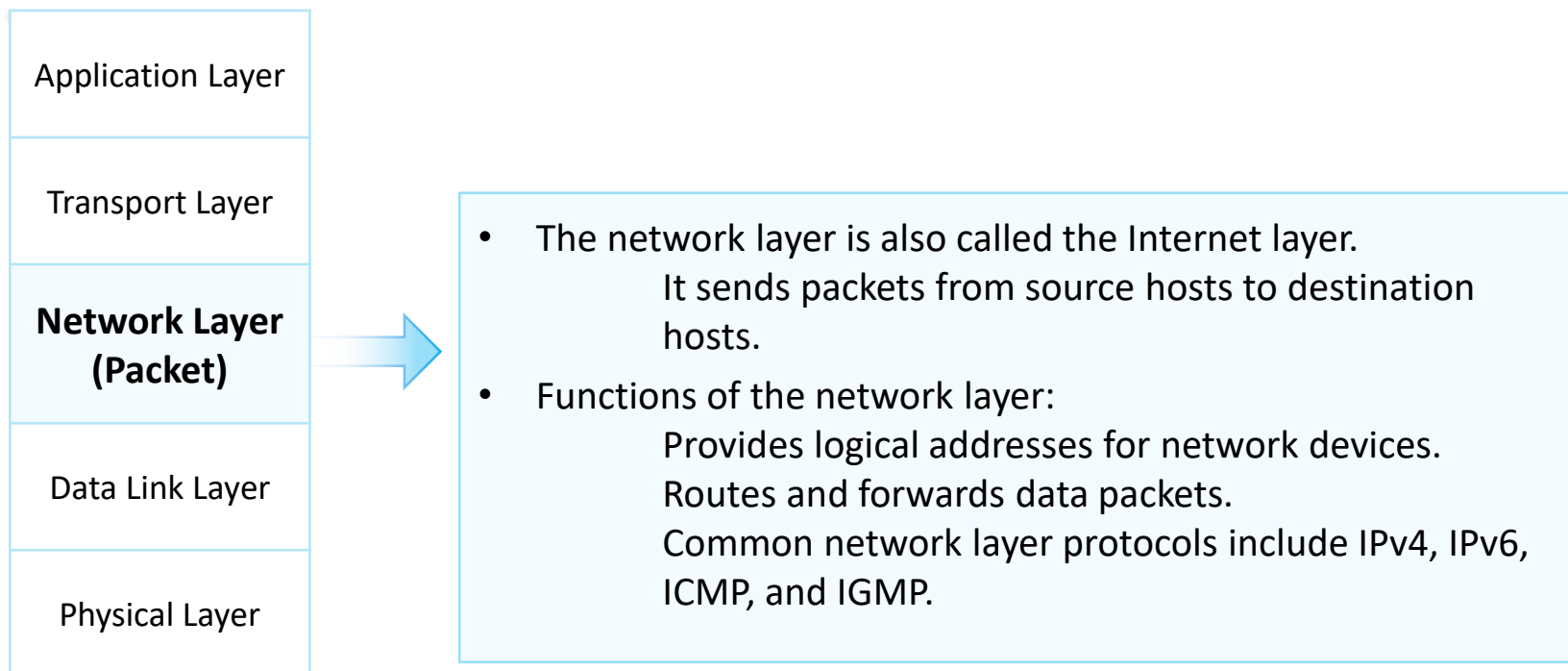
Transport
Layer

Network
Layer

Data Link
Layer

Physical
Layer

- The transport layer is responsible for establishing connections between processes on hosts, and the network layer is responsible for transmitting data from one host to another.
- PDUs transmitted at the network layer are called packets.





Working Process of a Network Layer Protocol

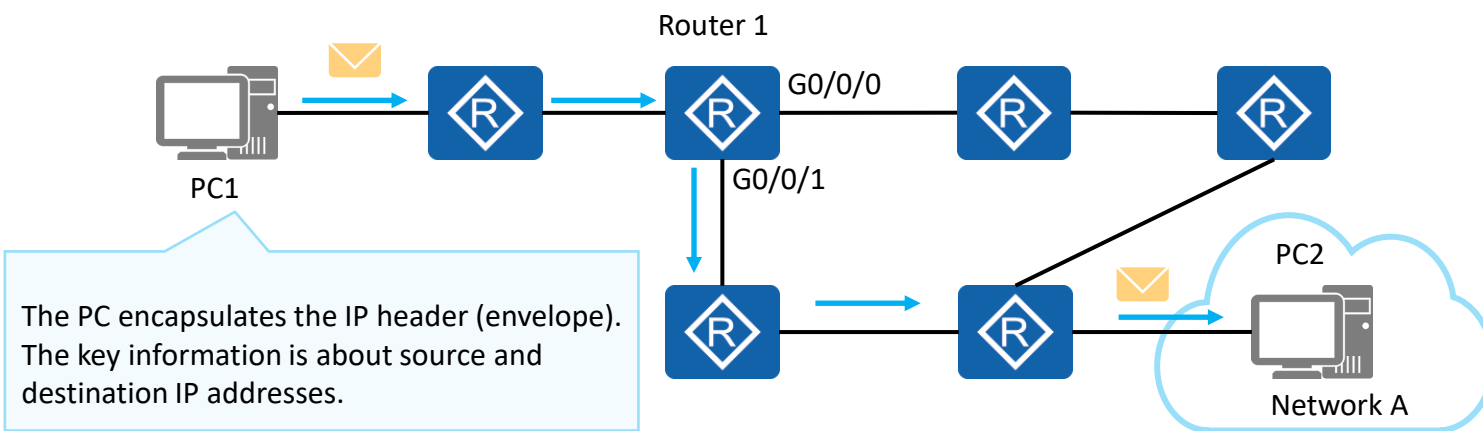
Packet Encapsulation

Letter: data sent by an upper layer (for example, the transport layer)



Envelope: IP packet header
Sender: source IP address
Receiver: destination IP address

Packet Forwarding Based on Network Layer Addresses



Routing table of Router 1

Network	Outbound Interface
Network A	G0/0/1
...	...
...	...

- The network layer header of a packet sent by a source device carries the network layer addresses of the source and destination devices.
- Each network device (such as a router) that has the routing function maintains a **routing table** (like a map of the network device).
- After receiving a packet, the network device reads the network layer **destination address** of the packet, searches the routing table for the matching entry of the destination address, and forwards the packet according to the instruction of the matching entry.



Data Link Layer

Application
Layer

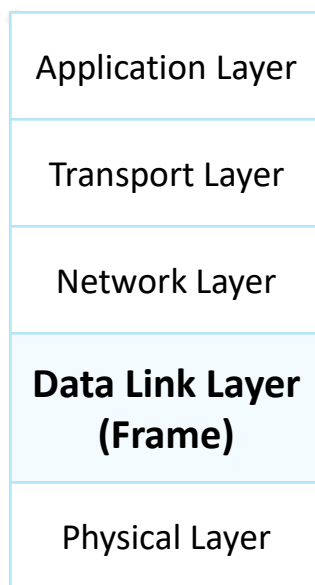
Transport
Layer

Network
Layer

Data Link
Layer

Physical
Layer

- The data link layer is located between the network layer and the physical layer and provides services for protocols such as IP and IPv6 at the network layer. PDUs transmitted at the data link layer are called frames.
- Ethernet is the most common data link layer protocol.



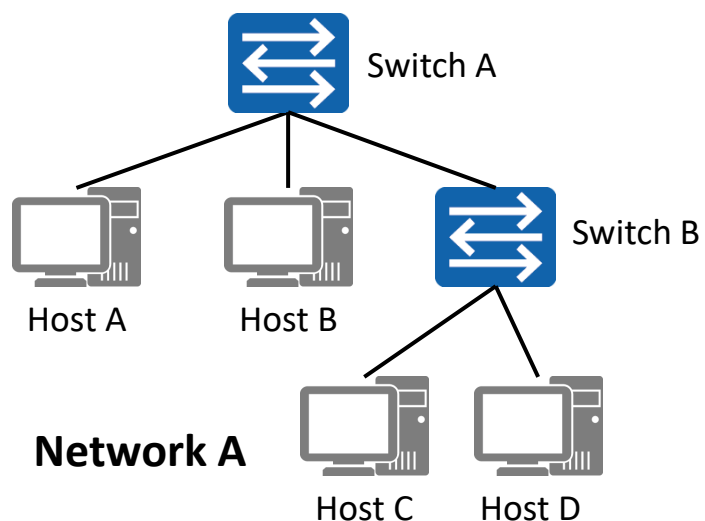
The data link layer is located between the network layer and the physical layer.

- The data link layer provides intra-segment communication for the network layer.
- The functions of the data link layer include framing, physical addressing, and error control.
- Common data link layer protocols include Ethernet, PPPoE, and PPP.



Ethernet and Source MAC Addresses

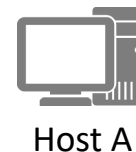
Ethernet Definition



- Ethernet is a broadcast multiple access protocol that works at the data link layer protocol.
- The network interfaces of PCs comply with the Ethernet standard.
- Generally, a broadcast domain corresponds to an IP network segment.

Ethernet Source MAC Addresses

I have a MAC address when I leave the factory.



Host A

Name: Host A

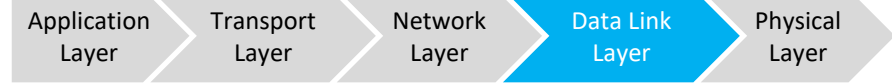


MAC address/Ethernet address/physical address:

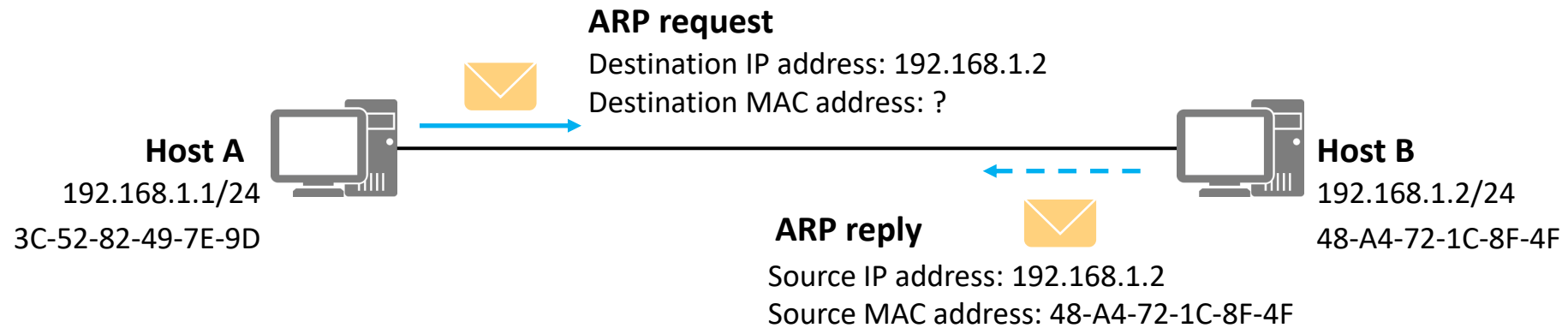
- A media access control (MAC) address uniquely identifies a NIC on a network. Each NIC requires and has a unique MAC address.
- MAC addresses are used to locate specific physical devices in an IP network segment.
- A device that works at the data link layer, such as an Ethernet switch, maintains a MAC address table to guide data frame forwarding.



ARP



- Address Resolution Protocol (ARP):
 - Discovers the MAC address associated with a given IP address.





ARP Implementation Principles (1)

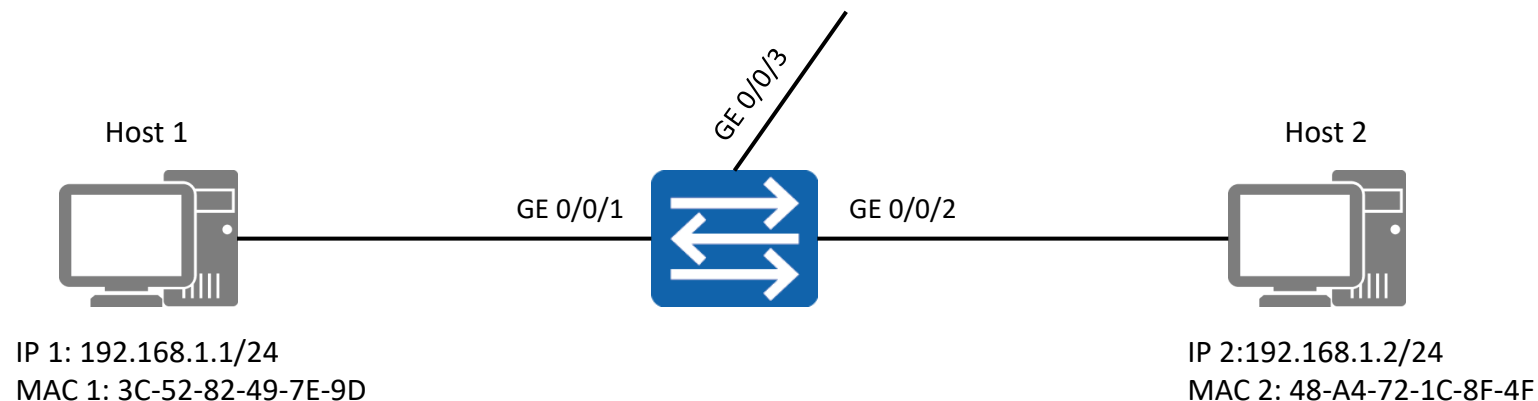
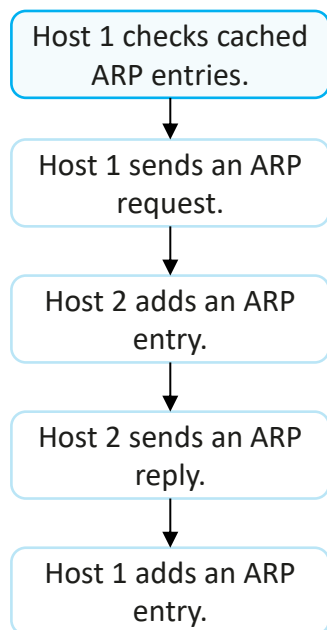
Application Layer

Transport Layer

Network Layer

Data Link Layer

Physical Layer



1

```
Host 1>arp -a
Internet Address  Physical Address  Type
```

The ARP cache table is empty.

Step 1:

- Before sending a datagram, a device searches its ARP table for the destination MAC address of the datagram.
- If the destination MAC address exists in the ARP table, the device encapsulates the MAC address in the frame and sends out the frame. If the destination MAC address does not exist in the ARP table, the device sends an ARP request to discover the MAC address.



ARP Implementation Principles (2)

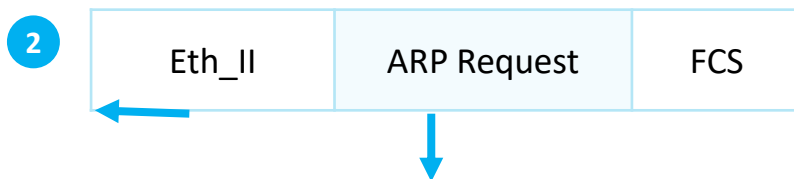
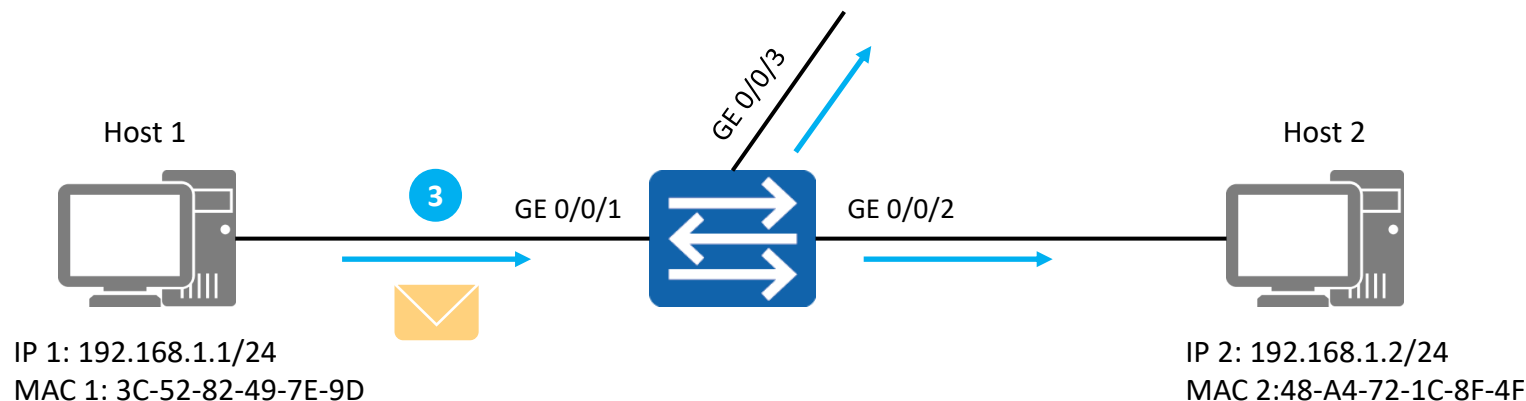
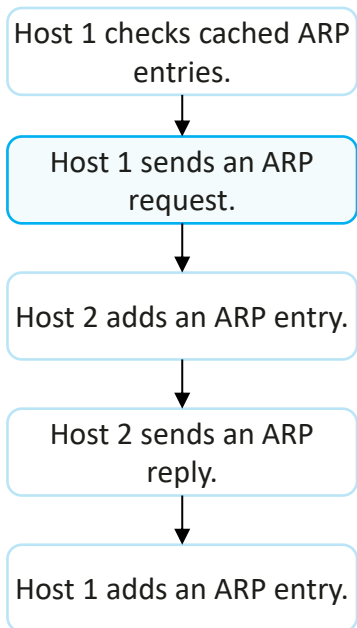
Application Layer

Transport Layer

Network Layer

Data Link Layer

Physical Layer



Destination MAC address:
FF-FF-FF-FF-FF-FF
Source MAC address:
MAC 1

Operation type: ARP request
MAC address of the sender: MAC 1
IP address of the sender: IP 1
Destination MAC address: 00-00-00-00-00-00
Destination IP address: IP 2

Step 2:

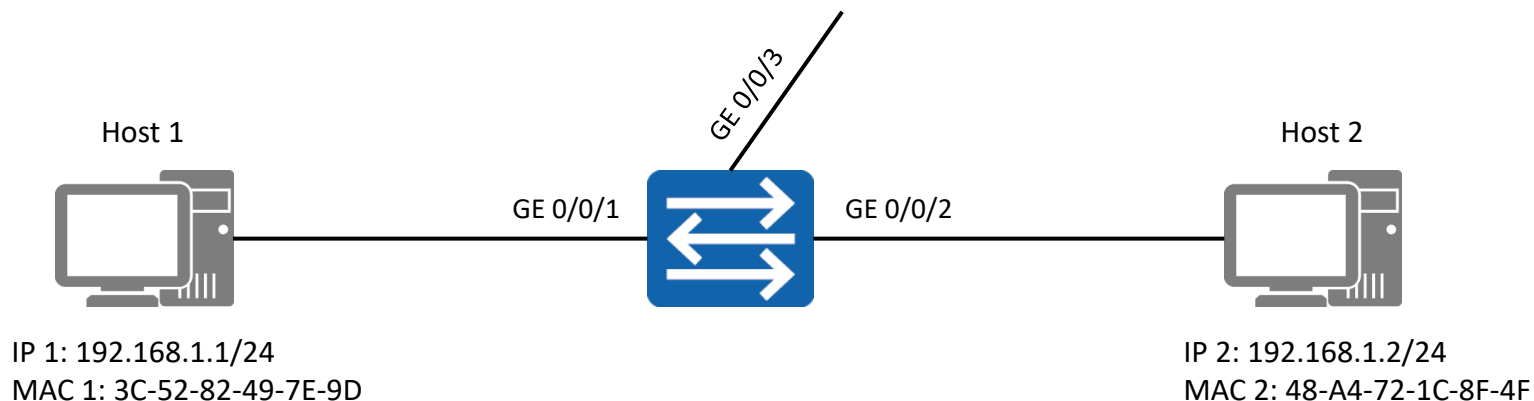
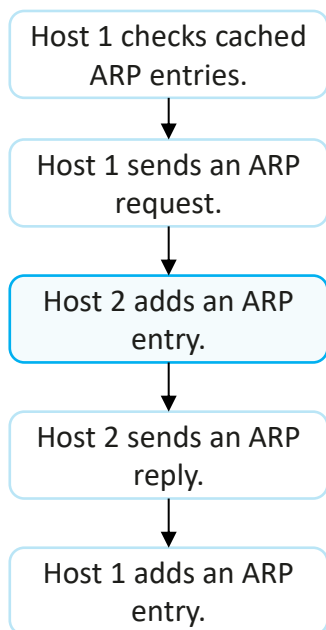
- Host 1 sends an ARP request to discover the MAC address of Host 2.
- The destination MAC address in the ARP request is 0 because the destination MAC address is unknown.

Step 3:

- The ARP request message is a broadcast data frame. After receiving the ARP request message, the switch floods it.



ARP Implementation Principles (3)



Step 4:

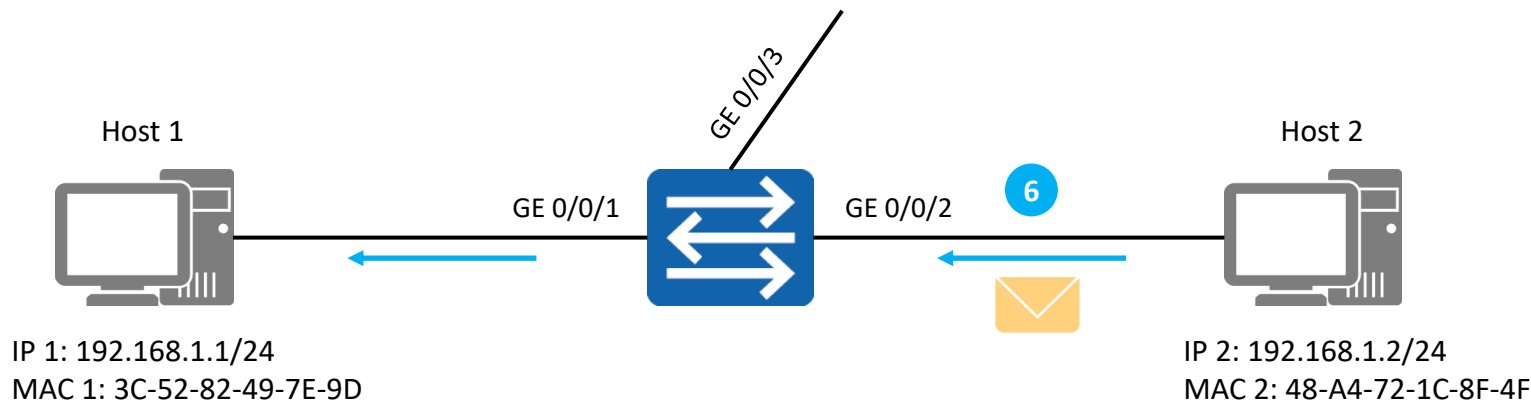
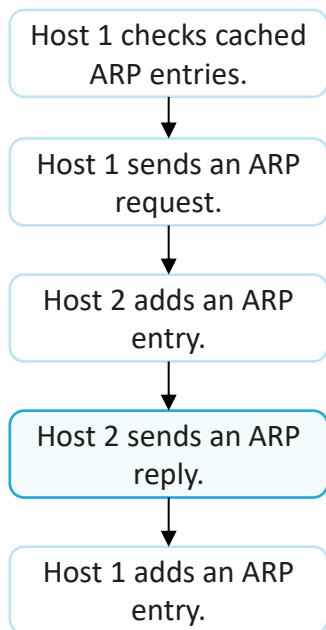
- After receiving the ARP request message, each host checks whether it is the destination of the message based on the carried destination IP address.
- Host 2 finds that it is the destination of the message and then records the mapping between the sender's MAC and IP addresses in its ARP table.

4

```
Host 2>arp -a
Internet Address      Physical Address      Type
192.168.1.1          3C-52-82-49-7E-9D    Dynamic
```



ARP Implementation Principles (4)



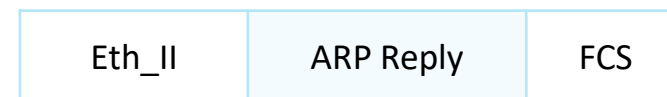
Step 5:

- Host 2 sends an ARP reply to Host 1.
- In this step, Host 2 has discovered the MAC address of Host 1, so the ARP reply is a unicast data frame.

Step 6:

- After receiving the unicast data frame, the switch forwards the frame.

5

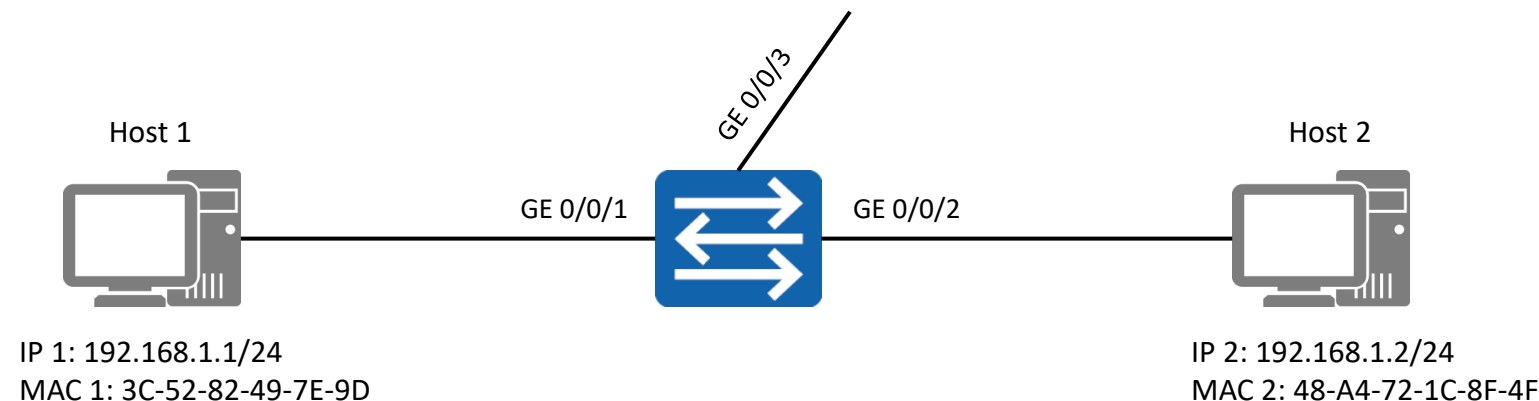
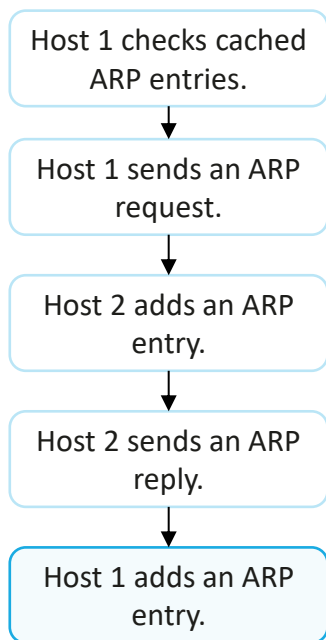


Destination MAC address: MAC-1
Source MAC address: MAC 2

Operation type: **ARP reply**
Sender's MAC address: **MAC 2**
Sender's IP address: **IP 2**
Receiver's MAC address: MAC 1
Receiver's IP address: IP 1



ARP Implementation Principles (5)



7

Internet Address	Physical Address	Type
192.168.1.2	48-A4-72-1C-8F-4F	Dynamic

Step 7:

- After receiving the ARP reply message, Host 1 checks whether it is the destination of the message based on the carried destination IP address.
- If so, Host 1 records the carried sender's MAC and IP addresses in its ARP table.



Physical Layer

- After data arrives at the physical layer, the physical layer converts a digital signal into an optical signal, an electrical signal, or an electromagnetic wave signal based on the physical media.
- PDUs transmitted at the physical layer are called bitstreams.



The physical layer is at the bottom of the model.

- This layer transmits bitstreams on media.
- It standardizes physical features such as cables, pins, voltages, and interfaces.
- Common transmission media include twisted pairs, optical fibers, and electromagnetic waves.



Common Transmission Media

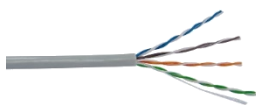
Application
Layer

Transport
Layer

Network
Layer

Data Link
Layer

Physical
Layer



Twisted pair



RJ45 connector

Data transmission through twisted pairs

1



Fiber



Optical module



Data transmission through optical fibers

2



Synchronous/asynchronous
serial cable: V.24 on the left and
V.35 on the right

Data transmission through serial cables

3

4



PAD



Mobile phone



Laptop



Wireless router

Data transmission between terminal and wireless
routers through wireless signals

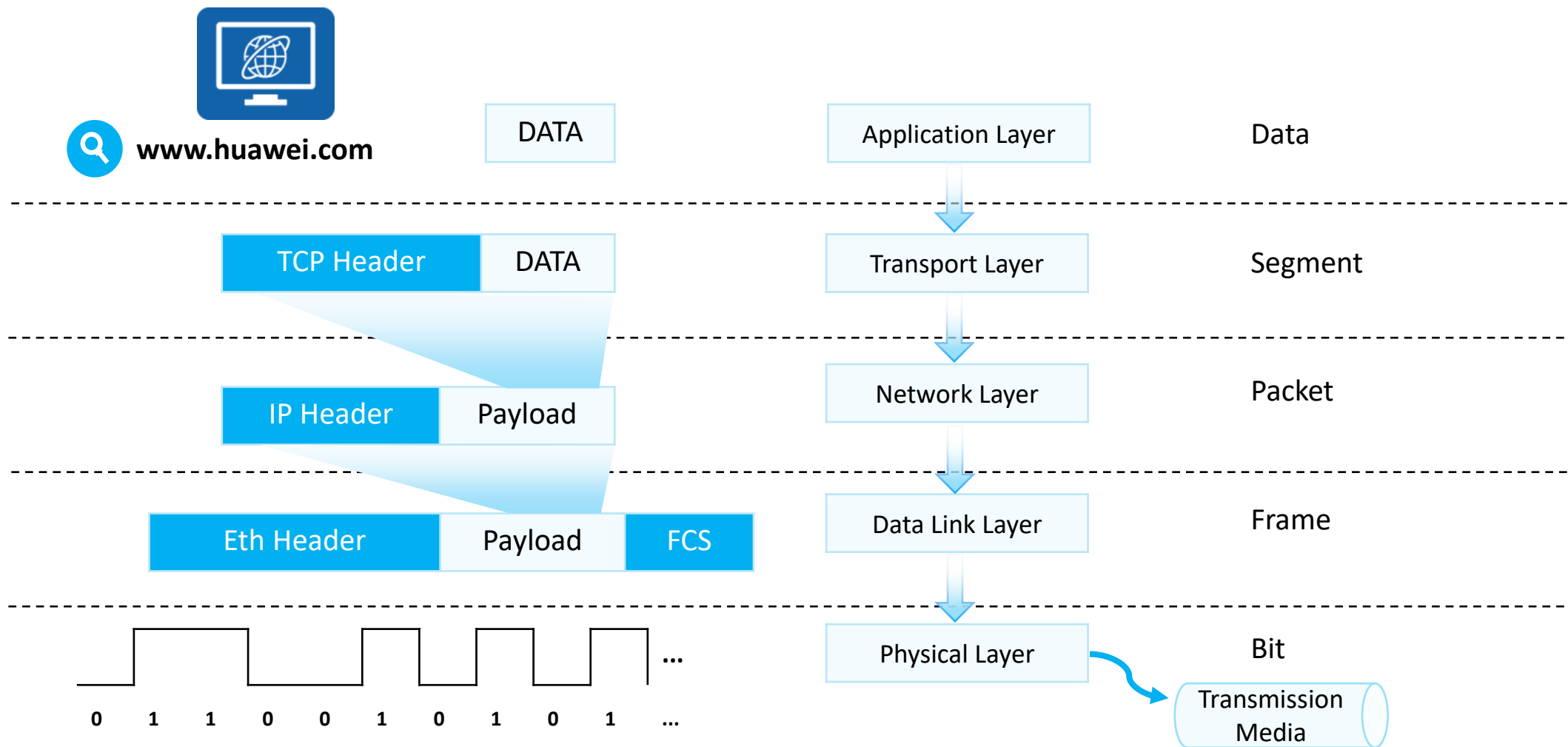


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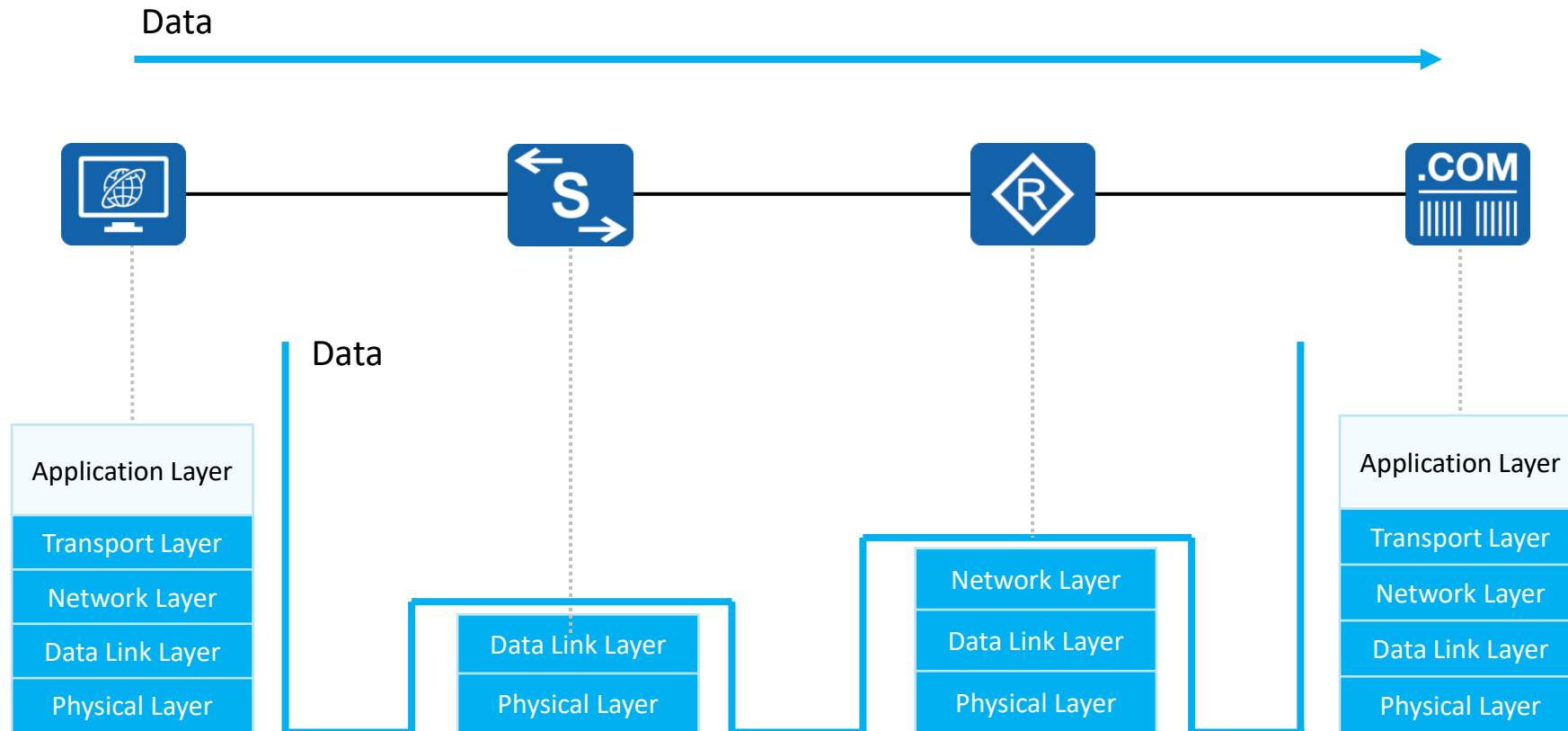
Data Encapsulation on the Sender





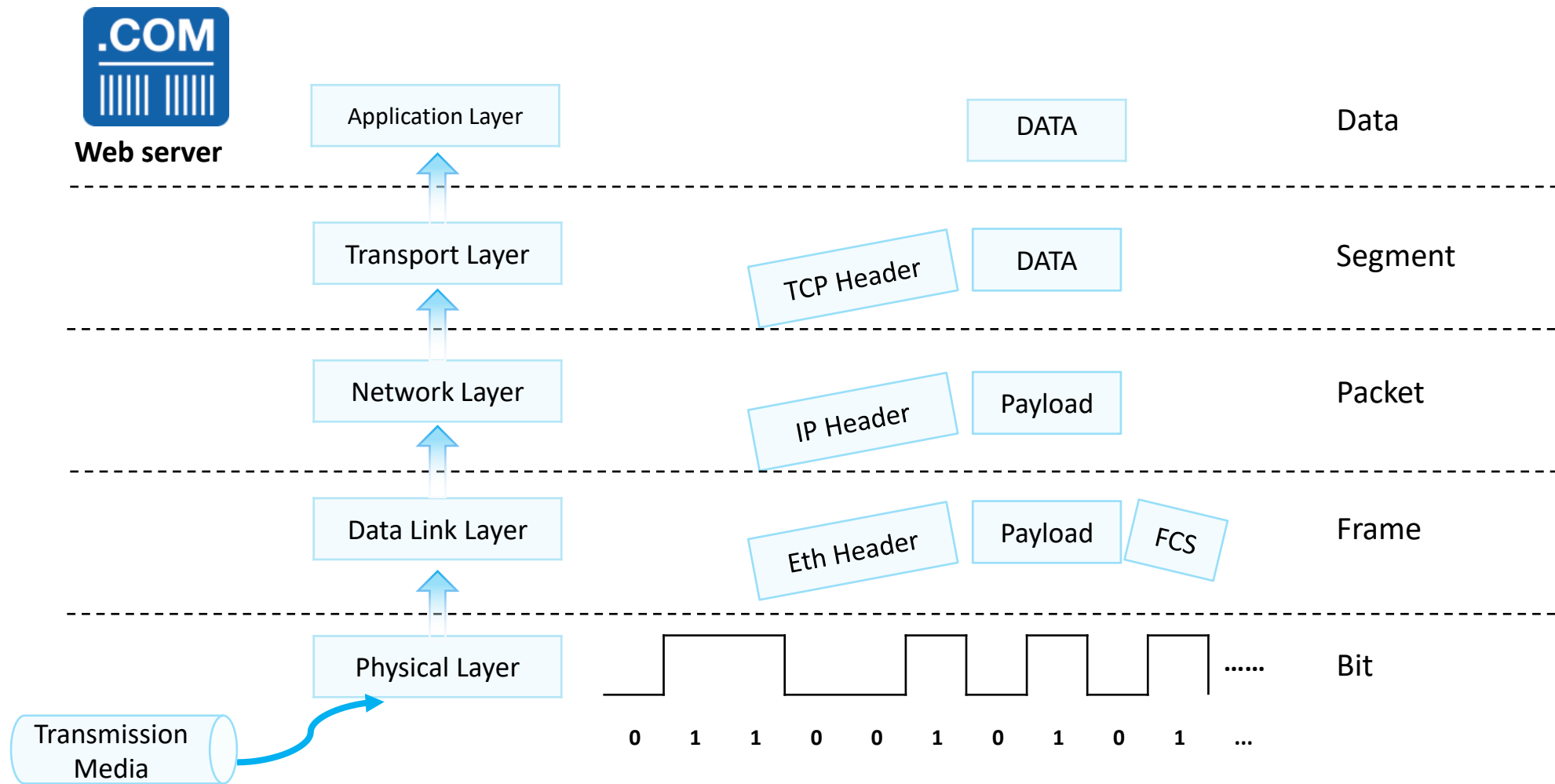
Data Transmission on the Intermediate Network

- Encapsulated data is transmitted on the network.





Data Decapsulation on the Receiver





Summary

- Both the OSI reference model and the TCP/IP reference model adopt the layered design concept.
 - Clear division of functions and boundaries between layers facilitates the development, design, and troubleshooting of each component.
 - The functions of each layer can be defined to impel industry standardization.
 - Interfaces can be provided to enable communication between hardware and software on various networks, improving compatibility.
- Data generation and transmission require collaboration between modules. Meanwhile, each module must fulfill its own responsibilities.



Quiz

1. What are the benefits of the layered model?
2. What are the common protocols at the application layer, transport layer, network layer, and data link layer?

The background of the image is a blue-tinted photograph of a modern office interior. In the foreground, several groups of business professionals are silhouetted against the bright light coming from large windows. They are engaged in various activities: some are looking at documents, others are in conversation, and one person is holding a folder. The office has a high ceiling and large glass panels. In the background, through the windows, a dense urban skyline with various skyscrapers is visible under a clear sky. The overall color palette is dominated by shades of blue, creating a professional and corporate atmosphere.

Thank You
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